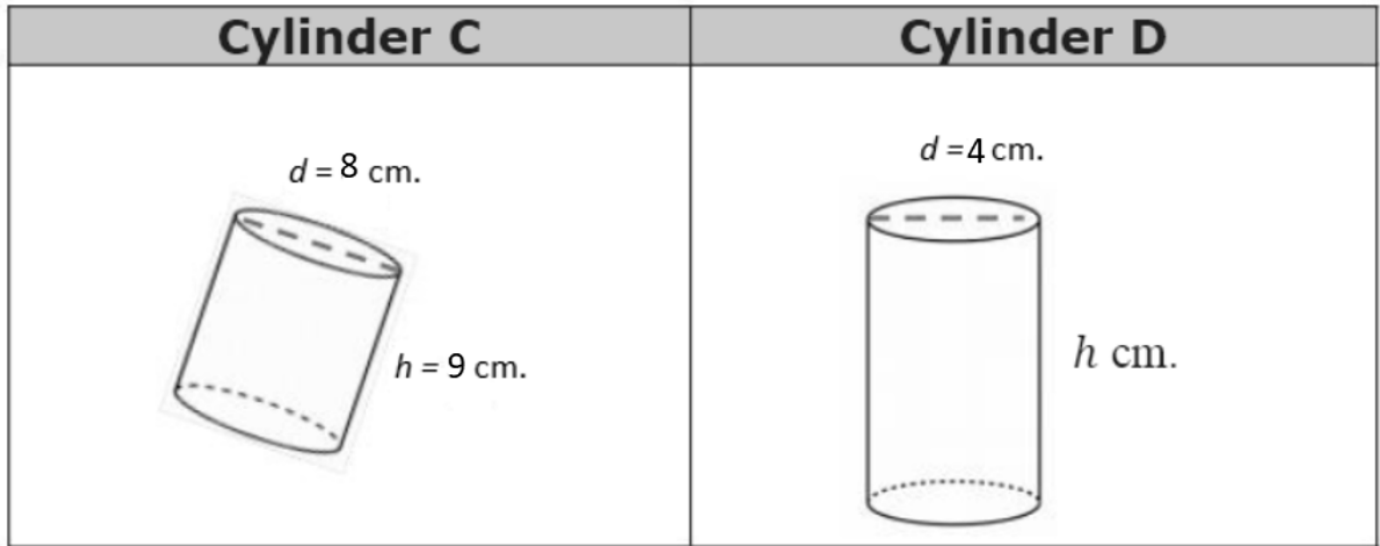
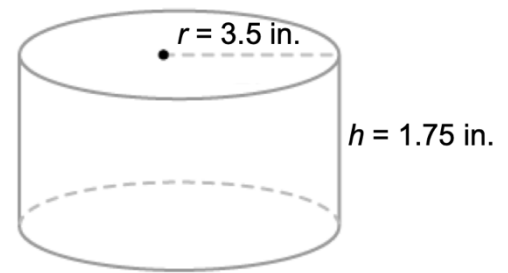


Cylinder C and Cylinder D have the same volume in cubic centimeters.



- Find the volume of Cylinder C . Leave it in terms of π .
- Write an expression for the volume of Cylinder D .
 $V =$ _____
- Write an equation to show the volumes are equal to each other.
- Solve the equation.
- What does the solution to the equation represent?

6. Find the volume of the cylinder shown. Write your answer in terms of π .



A bucket can hold 12,308.8 cubic inches. The diameter of the bucket is 28 inches (in.). What is the height of the bucket? Use 3.14 for π .

7. Sketch a diagram of the bucket and label the dimensions.

8. Find the height of the bucket.

9. The figure shows a coffee cup. The diameter measures 2 inches (in.) and the coffee cup can hold up to 16.49 cubic in. What is the height of the coffee cup? Use 3.14 for π .



Diameter: 2 in

10. A bucket in the shape of a cylinder is used to carry grain. The bucket has a diameter of 18 inches (in.) and can carry up to 6,104 cubic inches of grain. What is the height of the bucket? Use $\frac{22}{7}$ for π and round to the nearest hundredth.